

Lecture 3 – Energy Economics

Dr. Yu Zhang

ECE Department, UC Santa Cruz

ECE180J



Outline

- Efficiency and Capacity Factor
- Levelized Cost of Energy (LCOE, generation side)
- Electricity Rates (consumer side)

Efficiency & Capacity Factor

$$\text{Efficiency} = \frac{\text{actual energy output}}{\text{energy input}} \times 100\%$$

$$\text{Capacity Factor (CF)} = \frac{\text{actual energy output}}{\text{maximum possible energy output}} \times 100\%$$

Clearly, both quantities can be calculated by using “power” instead of “energy”.

Examples:

- A solar panel creates 300 kW of electricity from 900 kW of sunlight.
Its efficiency is $\frac{300 \text{ kW}}{900 \text{ kW}} \times 100\% = 33.3\%$.
- A power plant uses 3,412 MMBtu of fuels to generate 5×10^5 kWh energy in a week. Its efficiency is $\frac{5 \times 10^5 \text{ kWh}}{10^6 \text{ kWh}} \times 100\% = 50\%$ (1 kWh = 3,412 Btu).
- A 1MW wind turbine produces 3000MWh electricity in a year. Its capacity factor is $\frac{3000 \text{ MWh}}{1 \times 365 \times 24 \text{ MWh}} \times 100\% = 34.2\%$.

Heat Rate vs Efficiency

Key definitions

- Efficiency (η):
 $\eta = \text{Electrical Energy Output} / \text{Thermal Energy Input}$ (unitless, e.g., 0.4 = 40%)
- Heat Rate (HR):
 $\text{HR} = \text{Thermal Energy Input} / \text{Electrical Energy Output}$ (energy input per unit electricity)

Conversion formulas

- MJ/MWh: $\text{HR} = 3600/\eta$ (1 MWh = 3600 MJ)
- Btu/kWh: $\text{HR} = 3412/\eta$ (1 kWh = 3412 Btu)
- Consistent units: $\text{HR} = 1/\eta$ (holds when input/output use the same energy units)

Example ($\eta = 0.38$):

- $\text{HR} = 3600/0.38 \approx 9474 \text{ MJ/MWh}$
- $\text{HR} = 3412/0.38 \approx 8974 \text{ Btu/kWh}$

Time Value of Money

Q: You prefer to have \$10K today or ten years from now?

A: We could do many things with cash right now (rather not wait a decade):

Money is worth more in the present.

To account for the difference, we need to run a **present value analysis**: all future costs are converted into an equivalent present value:

Consider an investment of compound interest: future amount of money F in an account that starts with P , which earns annual interest i over a period of n years, will be:

$$F = P(1 + i)^n \implies P = \frac{F}{(1 + i)^n}$$

- When converting a future value F into a present value P , the interest term i is usually referred to as a discount rate d .
- The discount rate d reflects the opportunity cost of capital, which may differ from the borrowing interest rate i .

Discount Rate vs Interest Rate

Feature	Discount Rate (d)	Interest Rate (i)
Definition	Used to discount future cash flows to present value	Cost of borrowing or return on investment
Usage	Investment valuation, NPV calculations, monetary policy	Loans, bonds, mortgages, and savings accounts
Determined By	Businesses, investors, and central banks	Market conditions, central banks, and lenders
Effect	Higher d lowers present values	Higher i increases borrowing costs

Capital Recovery Factor (CRF): Annualizing Investment

A loan of P with interest rate i

A	A	A	A
0	1	2	$\dots$
			$n-1$
			n

A : annual payment

Task: Find the equal annual payment A for a n -year fixed-rate i loan P .

Solution: Evaluate the future value of P and all those A at a common time (e.g., the last year n)

$$\sum_{s=0}^{n-1} A(i+1)^s = A \frac{(i+1)^0 - (i+1)^n}{1 - (i+1)} = A \frac{(i+1)^n - 1}{i} = P(i+1)^n \Rightarrow$$

$$A(\$/\text{yr}) = P(\$) \cdot \text{CRF}(\%/ \text{yr}) \quad \text{where } \text{CRF} = \frac{i(1+i)^n}{[(1+i)^n - 1]}$$

The formula can be adjusted to find monthly payments

$$\text{CRF}(i, n) = \frac{(i/12)[1 + (i/12)]^{12n}}{[1 + (i/12)]^{12n} - 1} \text{per month}$$

Present Value Function (PVF)

Alternatively, all those future cash flows A can be discounted back to their present value using the specified discount rate i .

$$\frac{A}{1+i} + \frac{A}{(1+i)^2} + \cdots + \frac{A}{(1+i)^n} = P$$

This introduces another factor: **present value function (PVF)**:

$$\text{PVF}(d, n) = \frac{P}{A} = \sum_{s=1}^n \frac{1}{(1+d)^s} = \frac{(1+d)^n - 1}{d(1+d)^n}$$

Clearly, PVF is the reciprocal of CRF ($d=i$).

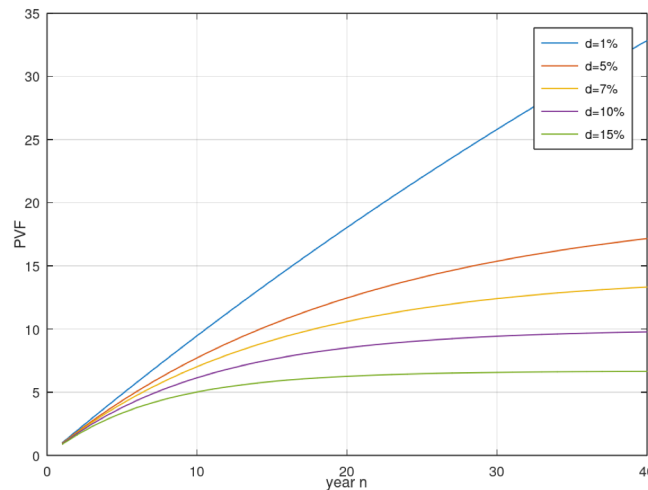


Fig: PVF as a function of n , evaluated for five discount rates d

Net Present Value (NPV)

NPV is difference between the discounted benefits and costs over time

$$\text{NPV}(i, n) = \sum_{t=0}^n \frac{B_t}{(1+i)^t} - \sum_{t=0}^n \frac{C_t}{(1+i)^t} = \sum_{t=0}^n \frac{R_t}{(1+i)^t}$$

where:

- B_t are the benefits or cash inflows at time t .
 - C_t are the costs or cash outflows at time t .
-

When a choice is to be made between two investments, the NPV of the savings can be found by

$$\text{NPV} = \Delta B \times \text{PVF}(d, n) - \Delta C$$

where:

- ΔB is the extra annual energy savings associated with the better product.
- ΔC is the extra capital cost of the better product.

Example: NPV of an Efficient Motor (1/2)

Consider two 100-hp electric motors. The cheap motor draws 79 kW and costs \$2400; the premium motor draws 75 kW and costs \$2900. The motors run 1600 h/yr with electricity costing \$0.08/kWh. Over a 20— year life, find the NPV of the premium motor if a discount rate is 10%.

Solution

The annual electricity savings for the premium motor is

$$\begin{aligned}\Delta B &= (79 - 75) \text{ kW} \times 1600 \text{ h/yr} \times \$0.08/\text{kWh} \\ &= \$512/\text{yr}\end{aligned}$$

The present value factor for a 20-year cash flow with a 10% discount rate is

$$\begin{aligned}\text{PVF}(d, n) &= \frac{(1 + d)^n - 1}{d(1 + d)^n} \\ &= \frac{(1 + 0.10)^{20} - 1}{0.10(1 + 0.10)^{20}} \\ &= 8.5136 \text{ yr}\end{aligned}$$

Example: NPV of an Efficient Motor (2/2)

The present value of 20 years of electricity savings for the premium motor is

$$PV(\text{savings}) = \Delta B \times PVF(d, n) = \$512/\text{yr} \times 8.5136\text{yr} = \$4359$$

The premium motor costs an additional \$500 so the NPV of the savings is

$$NPV = \$4359 - \$500 = \$3859$$

- If you buy the **cheap motor**, you pay an additional **\$512 per year** in electricity costs to the utility.
- If you buy the **premium motor**:
 - You take a **20-year loan at 10%** for **\$4359**.
 - You make **annual payments of \$512** to the bank (same as the extra electricity cost).
- Financial implication:
 - You only need **\$500 upfront** to cover the extra cost of the better motor.
 - You keep the remaining **\$3859 in cash immediately** (day one).
- Conclusion:
 - The **net present value (NPV)** is **\$3859**.

Equivalent Discount Rate with Fuel Escalation

Fuel costs may rise over time, increasing annual savings from efficiency measures. Let e denote the fuel escalation rate, then we have

$$\text{PVF}(d, e, n) = \frac{1+e}{1+d} + \frac{(1+e)^2}{(1+d)^2} + \dots + \left(\frac{1+e}{1+d} \right)^n$$

Then, we define the equivalent discount rate with fuel escalation as d'

$$\frac{1+e}{1+d} = \frac{1}{1+d'} \Rightarrow d' = \frac{d-e}{1+e}$$

$$\text{PVF}(d, e, n) = \text{PVF}(d', n) = \frac{(1+d')^n - 1}{d'(1+d')^n}$$

Example: PV Electricity Cost

A 3-kW PV system, which operates with a capacity factor (CF) of 0.25, costs \$10,000 to install. There are no annual costs associated with the system other than the payments on a 6%, 20-year loan. Find the cost of electricity generated by the system (¢/kWh).

Solution:

$$A = P \times \text{CRF}(0.06, 20) = \$10,000 \times 0.0872/\text{yr} = \$872/\text{yr}$$

$$\text{Annual energy (kWh/yr)} = 3 \text{ kW} \times 8760 \text{ h/yr} \times 0.25 = 6570 \text{ kWh/yr}$$

The cost of electricity from the PV system is therefore

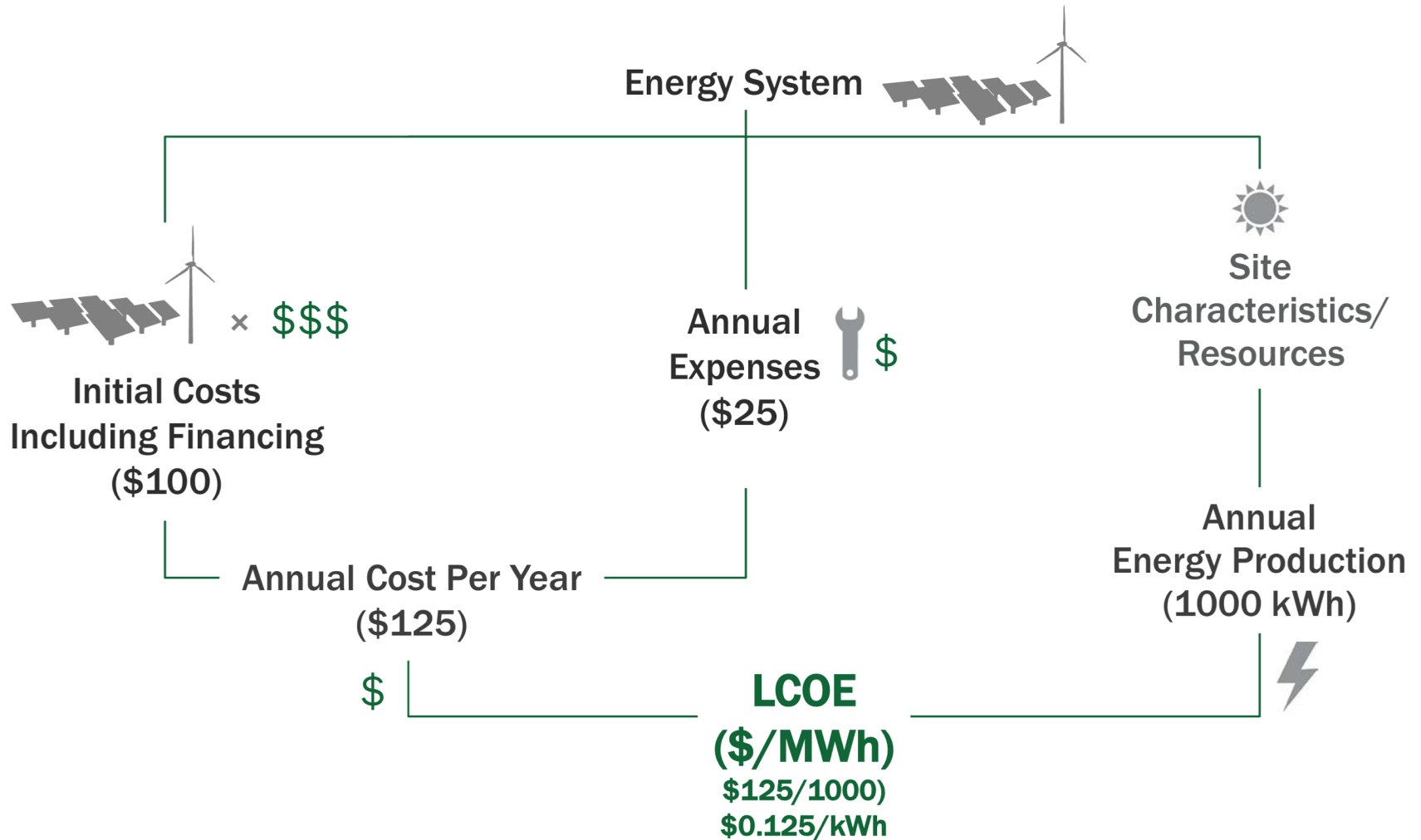
$$\text{Cost of PV electricity} = \frac{\$872/\text{yr}}{6570 \text{ kWh/yr}} = \$0.133/\text{kWh} = 13.3\text{¢}/\text{kWh}$$

Levelized Cost of Energy (LCOE)

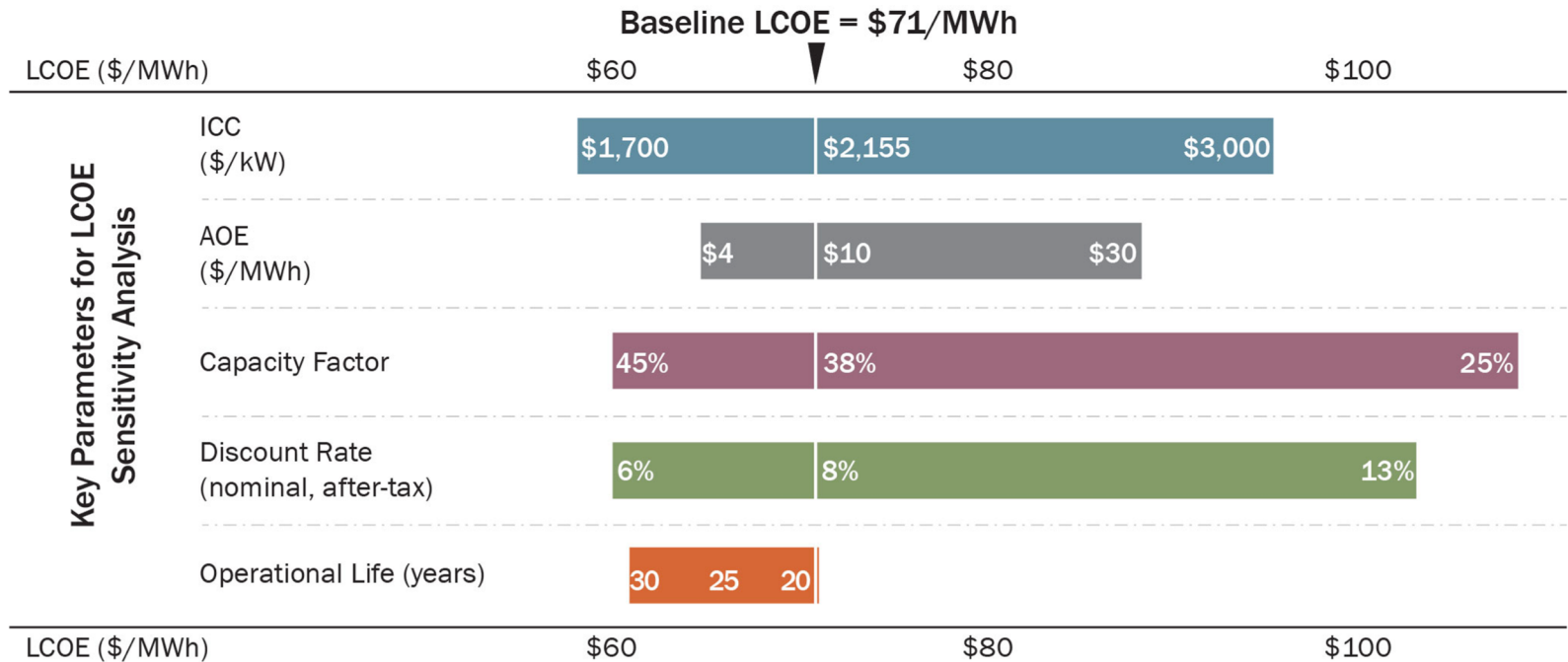
$$\text{LCOE (\$/kWh)} = \frac{[\text{Annual fixed cost} + \text{Annual variable cost}] (\$/\text{yr})}{\text{Annual output (kWh/yr)}}$$

- “Levelized” refers to averaging the costs over time while taking the time value of money into account: It spreads out all the costs and outputs **over the system’s lifetime** in a way that allows fair comparison across different technologies.
- LCOE provides a way to assess the cost of a project, factoring **in initial investment, operational costs, financing, and other relevant expenses**. Those inputs to LCOE are chosen by the estimator.
- LCOE does not directly determine retail electricity prices, but it is widely used to compare generation technologies: A useful tool for **policymakers, investors, and energy planners** to assess the economic competitiveness of various energy technologies.

LCOE Concept



Big Driving Factors of Wind LCOE



Source: Tegen et al. 2012

- Initial capital cost (ICC) and capacity factor are two critical drivers.
- Discount rate (financing costs) and annual operating expenses (AOE) are non-trivial.

Levelized Fixed and Variable Costs

- **Fixed costs** include capital costs, taxes, insurance, and any fixed O&M costs that will be incurred even when the plant isn't operated.

$$\text{Levelized fixed cost (\$/kWh)} = \frac{\text{Capital cost (\$/kW)} \times \text{FCR(1/yr)}}{8760 \text{ h/yr} \times \text{Capacity factor}}$$

where Fixed Charge Rate (FCR) accounts for interest on loans, acceptable returns for investors, fixed O&M, charges, taxes, etc.

- **Variable costs** are the added costs associated with actually running the plant. These are mostly fuel plus variable O&M costs.

$$\begin{aligned} &\text{Levelized annual cost (\$/kWh)} \\ &= \left[\text{Heat rate} \left(\frac{\text{Btu}}{\text{kWh}} \right) \times \text{Fuel} \left(\frac{\$}{\text{Btu}} \right) + \text{O\&M} \left(\frac{\$}{\text{kWh}} \right) \right] \times \text{LF} \end{aligned}$$

where:

- Levelizing Factor (LF) is a multiplier that converts the escalating annual fuel and O&M costs into a series of equal annual amounts.

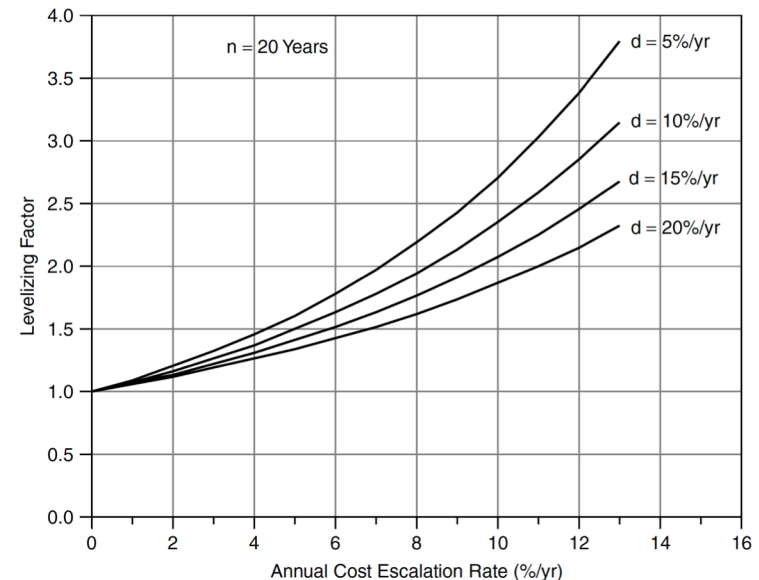
Levelizing Factor (LF)

- To account for the cost escalation, inflation (fuel price, variable O&M etc), we need to multiply the variable costs with the levelizing factor (LF).

$$\text{Levelizing factor (LF)} = \underbrace{\left[\frac{(1 + d')^n - 1}{d'(1 + d')^n} \right]}_{\text{PVF}(d', n)} \cdot \underbrace{\left[\frac{d(1 + d)^n}{(1 + d)^n - 1} \right]}_{\text{CRF}(d, n)}$$
$$d' = \frac{d - e}{1 + e}$$

if $e = 0 \implies$
 $d' = d, \text{ LF} = 1$

- By multiplying the PVF and CRF, we essentially incorporate both the time value of money (discounting future cash flows to present value) and the recovery of capital costs over time (annualized payments to cover initial investment and return).



Find Annual Fixed Cost

Example 1. Consider a natural-gas fired combined cycle (NGCC) power plant with a total installed cost of \$1300/kW. Assume this is an investor-owned utility (IOU) with 52% equity financing at 11.85% and 48% debt at 5.40% with investments "booked" on a 20-year term. Add 2% of the capital cost per year to account for insurance, property taxes, fixed O&M, and another 4% for corporate taxes. Find the annual fixed cost of this plant (\$/yr-kW).

Solution.

Step 1) find the weighted average cost of capital:

$$\text{Average cost of capital} = 0.52 \times 11.85\% + 0.48 \times 5.40\% = 8.754\%$$

Step 2) compute the CRF:
$$\text{CRF} = \frac{0.08754(1 + 0.08754)^{20}}{[(1 + 0.08754)^{20} - 1]} = 0.10763/\text{yr} = 10.763\%/\text{yr}$$

Step 3) compute the FCR:

$$\text{FCR} = 10.763\% \text{ (finance)} + 2\% \text{ (fixed O\&M, etc.)} + 4\% \text{ (taxes)} = 16.763\%$$

Step 4) compute the annual fixed cost:

$$\text{Annual fixed cost} = \$1300/\text{kW} \times 0.16763/\text{yr} = \$218/\text{yr-KW}$$

Find LCOE

Example 2. Consider the NGCC plant in Example 1. Suppose natural gas now costs \$6/MMBtu and it is projected to rise at 5%/yr in the future. The owners have a 10% discount factor. Annual O&M adds another 0.4 ¢/kWh. If its heat rate is 6900 Btu/kWh and the plant has a 70% CF, find its LCOE.

Solution. Using Equation 1.9 with an assumed 1 kW of rated power, the annual energy delivered per kW of rated power is

$$\text{Annual energy} = 1 \text{ kW} \times 8760 \text{ h/yr} \times 0.70 = 6132 \text{ kWh/yr}$$

From Figure 1.28 the levelizing factor for fuel is 1.5. From Equation 1.10, the annualized fuel cost per kW is

$$\begin{aligned} \text{Annual fuel cost (per kW)} &= 6132 \text{ kWh/yr} \times 6900 \text{ Btu/kWh} \times \$6/10^6 \text{ Btu} \times 1.5 \\ &= \$381/\text{yr} \end{aligned}$$

LCOE (cont'd)

$$\text{Annual energy} = 1 \text{ kW} \times 8760 \text{ h/yr} \times 0.70 = 6132 \text{ kWh/yr}$$

$$\text{Annual O\&M adds another } \$0.004/\text{kWh} \times 6132 \text{ kWh/yr} = \$25/\text{yr}$$

$$\text{Total variable costs (per kW)} = \$381 + \$25 = \$406/\text{yr}$$

Adding the \$218/yr-kW for annualized fixed costs gives a total

$$\text{Total annualized costs} = (\$218 + \$406) \text{ \$/yr-kW}$$

Using Equation 1.12, the total levelized cost is therefore

$$\text{LCOE} = \frac{\$218/\text{yr-kW} + \$406/\text{yr-kW}}{8760 \text{ h/yr} \times 0.70} = \$0.1017/\text{kWh} = 10.17\text{¢}/\text{kWh}$$

Alternative Formula for LCOE

- LCOE is a measure of the average net present cost of electricity generation over the lifetime of a power plant.

$$\text{LCOE} = \frac{\text{sum of costs over lifetime}}{\text{sum of energy produced over lifetime}} = \frac{\sum_{t=1}^n \frac{I_t + M_t + F_t}{(1+d)^t}}{\sum_{t=1}^n \frac{E_t}{(1+d)^t}}$$

- I_t : investment expenditures in year t
- M_t : operations and maintenance expenditures in year t
- F_t : fuel expenditures in year t
- E_t : electrical energy generated in year t
- d : discount rate
- n : expected lifetime of the power plant

LCOE Comparison (Utility-scale, Typical Ranges)

Energy Source	LCOE (¢/kWh)	Key Cost Drivers	Notes
Solar PV (utility)	2 – 6	Capital cost, capacity factor	Lowest in sunny regions
Onshore Wind	3 – 7	Wind resource, turbine cost	Site-dependent
Offshore Wind	7 – 14	Installation, maintenance, transmission	Higher infrastructure cost
Hydropower	3 – 10	Site-specific infrastructure	Long lifetime
Geothermal	5 – 10	Drilling cost, resource quality	Stable baseload
Biomass	6 – 12	Fuel supply cost	Dispatchable renewable
Natural Gas (Combined Cycle Gas Turbine, CCGT)	4 – 9	Fuel price, efficiency	Flexible generation
Coal	6 – 15	Fuel, emissions controls	Declining usage
Nuclear	12 – 20	High capital, financing	Low operating cost

Electricity Rates

Energy Statement



ENERGY STATEMENT

www.pge.com/MyEnergy

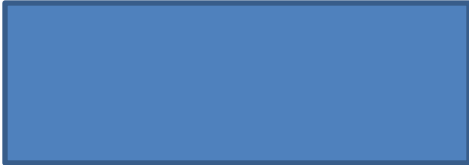
1

Account No: 1023456789-0

Statement Date: 02/19/2019

Due Date: 03/12/2019

Service For:



Questions about your bill?

24 hours per day, 7 days per week

Phone: 1-866-743-0335

www.pge.com/MyEnergy

Local Office Address

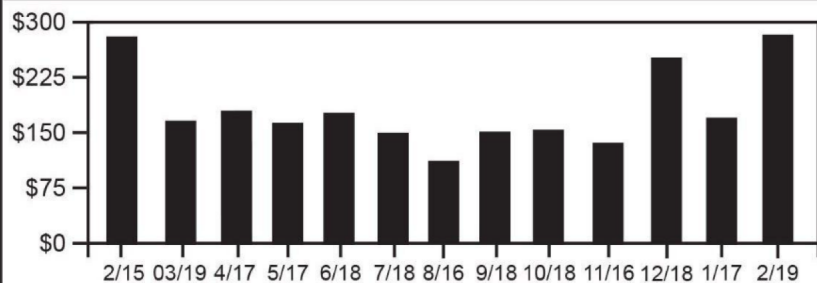
100 N. MAIN ST
FORT BRAGG, CA 95437

Your Account Summary

Credit Balance on Previous Statement	\$179.42
Payment(s) Received Since Last Statement	-179.42
Previous Unpaid Balance	0.00
2 Current PG&E Electric Delivery Charges	\$185.01
3 Sonoma Clean Power Electric Generation Charges	90.54

Total Amount Due by 03/12/2019 \$275.55

Electric Monthly Billing History

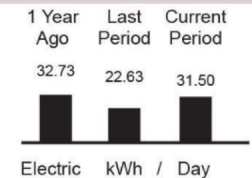


2018

Visit www.pge.com/MyEnergy for a detailed bill comparison

2019

Daily Usage Comparison



Energy Statement (Cont'd)



ENERGY STATEMENT

www.pge.com/MyEnergy

Account No: 1023456789-0
Statement Date: 02/19/2019
Due Date: 03/12/2019

Details of PG&E Electric Delivery Charges

01/11/2019 - 02/21/2019 (32 billing days)

Service For: 281 TALON DR

Service Agreement ID: 2416854841

Rate Schedule: E1 X Residential Service

Service Information

Meter # 1005868421
Current Meter Reading 30,877
Prior Meter Reading 29,869
Total Usage 1,008.000000 kWh
Baseline Territory X
Heat Source B - Not Electric
Serial Q
Rotating Outage Block 12P

01/11/2019 - 02/11/2019

Your Tier Usage

1

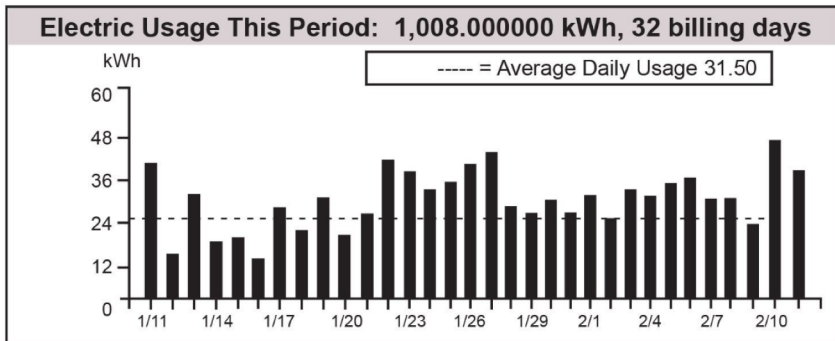
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Tier 1 Allowance	342.40 kWh (32 days x 10.7 kWh/day)	
Tier 1 Usage	342.400000 kWh @ \$0.21183	\$72.53
Tier 2 Usage	665.600000 kWh @ \$0.28011	186.44
Generation Credit		-108.66
Power Charge Indifference Adjustment		34.15
Franchise Fee Surcharge		0.55

Total PG&E Electric Delivery Charges

\$185.01

2013 Vintaged Power Charge Indifference Adjustment



Different Charges

Most common components in utility bills:

1) Energy charges

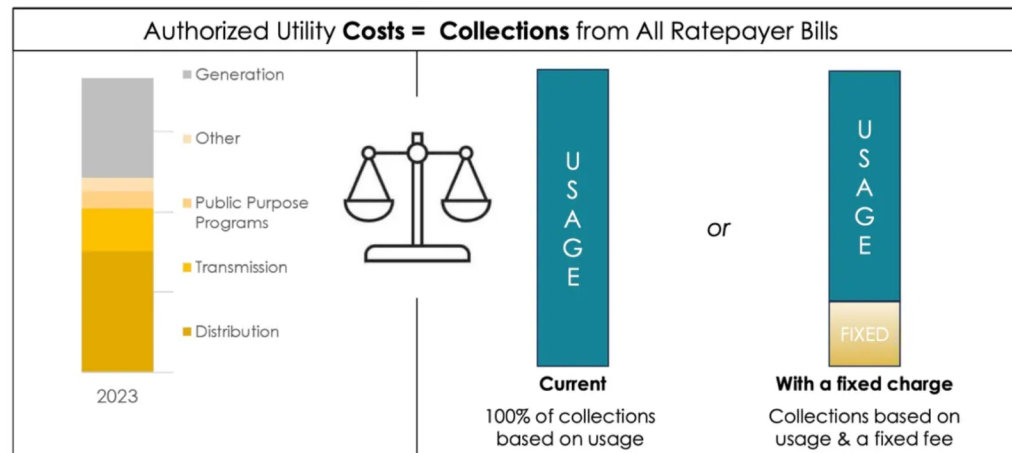
- Based on the amount of electricity (kWh) consumed.
- Cost can vary by time of use and by season.

2) Demand charges

- Based on maximum demand (kW) in each month.
- Shifting usage from high to low demand period lowers this charge.

3) Fixed charges

- Includes maintenance, generation, T&D, wildfire mitigation and public programs (e.g., California Alternate Rates for Energy (CARE)).
- Billed monthly and determined by rate schedule, not consumption.



(CPUC Public Advocates Office)

Energy Charge

- You pay for the total amount of electrical energy that is used, typically measured in kilowatt-hours (kWh).

Example: At a rate of 10 cents per kWh, how much does it cost to keep a 100 W light bulb on for one day?

Solution: 100 W = 0.1 kW, 1day = 24 hours, so

$$\text{cost} = 0.1 \text{ kW} \times 24 \text{ hours} \times \$0.10/\text{kWh} = \$0.24 = 24 \text{ ¢}$$

→ for one month (30days) that amounts to \$7.20

Demand Charge

$$\text{Load factor (\%)} = \frac{\text{Average power}}{\text{Peak power}} \times 100\%$$

❑ Load factor impact:

- Customer A with a peak demand of 100 kW and annual energy consumption of 876,000 kWh has a load factor of 100%.
- Customer B with the same annual energy consumption of 876,000 kWh, but a peak demand of 200 kW has a load factor of 50%.

❑ Utility capacity requirements:

- The utility needs twice as much generation capacity and transmission and distribution infrastructure to serve a customer with a lower load factor.
- Higher capacity requirements increase costs for the utility and necessitate rate adjustments to recover additional expenses.

❑ Role of rate structure:

- Energy charges (kWh) remain the same for both customers since they use the same amount of energy annually.
- Demand charges (kW-mo) vary based on peak demand, reflecting the capacity needed to serve customers during periods of high demand.

An Example

Two customers each use 100,000 kWh/mo. Customer A has a load factor of 15% and Customer B has a 60% load factor. Assume energy charges of \$0.06/kWh and demand charges of \$10/kW-mo, compare their monthly utility bills.

Solution: They both have the same energy costs:

$$100,000 \text{ kWh/mo} \times \$0.06/\text{kWh} = \$6000/\text{mo}$$

$$\text{Peak demand(A)} = \frac{100,000 \text{ kWh/mo}}{15\% \times 24 \text{ h/day} \times 30 \text{ day/mo}} = 925.9 \text{ kW, costing } \$9259/\text{mo}$$

$$\text{Peak demand(B)} = \frac{100,000 \text{ kWh/mo}}{60\% \times 24 \text{ h/day} \times 30 \text{ day/mo}} = 231.5 \text{ kW, costing } \$2315/\text{mo}$$

Fixed Charge

Proposed Fixed Charge					
CARE	Income Bracket	Annual Income	PG&E	SCE	SDG&E
Non-CARE		< \$50,000	\$0.00 (\$22.79 without the CCC)	\$0.00 (\$21.82 without the CCC)	\$0.00 (\$26.70 without the CCC)
	1				
	2	\$50,000 - \$100,000	\$31.91	\$30.55	\$36.42
	3	> \$100,000	\$36.69	\$35.14	\$41.88
CARE		< \$50,000	\$0.00 (\$10.20 without the CCC)	\$0.00 (\$10.83 without the CCC)	\$0.00 (\$13.70 without the CCC)
	1				
	2	\$50,000 - \$100,000	\$14.27	\$15.17	\$19.18
	3	> \$100,000	\$16.41	\$17.44	\$22.06

The Public Advocates Office

Public Advocates Office proposal for income-based fixed charges for California's three big investor-owned utilities (Public Advocates Office)

1. <https://www.publicadvocates.cpuc.ca.gov/press-room/reports-and-analyses/income-graduated-fixed-charge-qa>

2. <https://www.canarymedia.com/articles/utilities/bill-would-end-california-experiment-with-income-based-electric-bills>

How Electricity Rate is Decided

The factors deciding the electricity rate

- electric distribution and transmission
- costs of procuring power
- revenues needed for public programs (e.g., low-income and energy-efficiency programs).
- an approved rate of return for shareholder investments

Rates are the same across PG&E's entire service area. However, the amount of electricity allotted at the lowest available price (baseline) is different for each customer.

Formulas for Calculating Electricity Rate

$$TRR = OE + r \times (RB)$$

Where:

TRR = total revenue requirement (\$)

OE = operating expenses (\$)

r = rate of return on rate base (%)

RB = rate base (\$)

- FERC has jurisdiction over electric transmission rate base and rates.
- CPUC has jurisdiction over generation and distribution grid.

$$\frac{TRR_e \times p}{S \times s} = ARR_e$$

Where: TRR_e = total electric revenue requirement (\$) for a given year

p = residential revenues as % of total revenue (%) calculated using historic data

S = total sales (kWh) forecasted for that same year

s = residential sales as % of total sales (%)

ARR_e = average residential electric rate (\$/kWh) for that year

Regulation of Electricity Rate

- When PG&E makes any rate change, they need to submit an application to the California Public Utilities Commission (CPUC).
- The application is reviewed in a public hearing and by stakeholders (such as groups representing residential and business consumers, low-income and community advocates, environmental groups, agricultural interests and others).
- After this lengthy process, the CPUC issues a decision. PG&E then incorporates the approved changes into rates.

PG&E changes its electricity rates only **2-3 times a year**.
Their natural gas supply rates change **every month**.

Types of Rate Plans

Three main rate plans by PG&E: 1) Tiered rate plan; 2) Time-of-use rate plan; 3) Plug-in Electric Vehicles rate plan

Tiered Rate Plan (E-1)

Tier 1
Up to 100%
of Baseline
Allocation

31¢

Tier 2
101-400%
of Baseline
Allocation

39¢

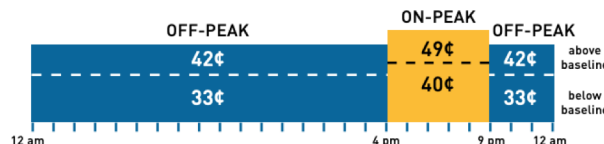
High Usage
>400%
of Baseline
Allocation

49¢

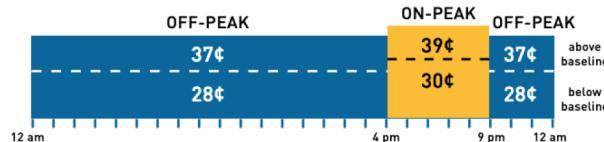
Time-of-Use Rate Plans

Time-of-Use (Peak Pricing 4-9 p.m. Every Day) (E-TOU-C)

Summer Season
June 1-Sept 30

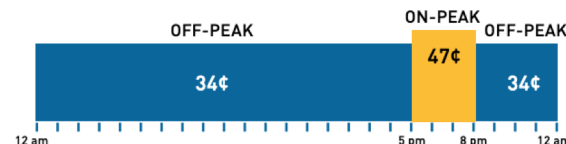


Winter Season
Oct 1-May 31

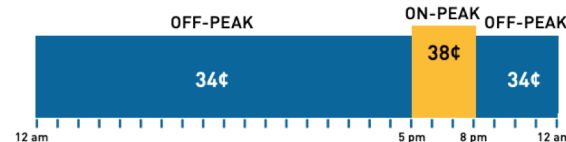


Time-of-Use (Peak Pricing 5-8 p.m. Weekdays) (E-TOU-D)

Summer Season
June 1-Sept 30



Winter Season
Oct 1-May 31



E-TOU-D may be more attractive for higher energy users

Baseline Allowance and Add-on

Baseline Allowance consists of an allotment of energy available at the lowest price

- It is the minimum level of usage needed to satisfy a substantial portion of the electricity needs of the average customer in a specific geographic region called a **Climate Zone**.
- Zones don't follow county/city boundaries, but are based on areas with similar geography and climate, the season, and the household heating source.
- Customers in hotter Climate Zones have higher.
- PG&E has 16 different Climate Zones.

There are several **add-ons** added to the base plan, e.g., SmartRate Add-on, Net-energy metering Add-on, and Solar Choice plans

1. https://www.pge.com/en_US/residential/rate-plans/how-rates-work/learn-how-rates-are-set/learn-how-rates-are-set.page

2. https://www.pge.com/en_US/residential/rate-plans/rate-plan-options/understanding-rate-plans/understanding-rate-plans.page

Problem: California's High Electricity Rates Crisis



Avg Residential Rate
(\$ per kWh, Jan 2024)

\$0.396

10-Year Change
Jan 2014 to Jan 2024

↑ 127%

**% of Customers
Behind on Bills**

22%



\$0.334

↑ 91%

19%



\$0.362

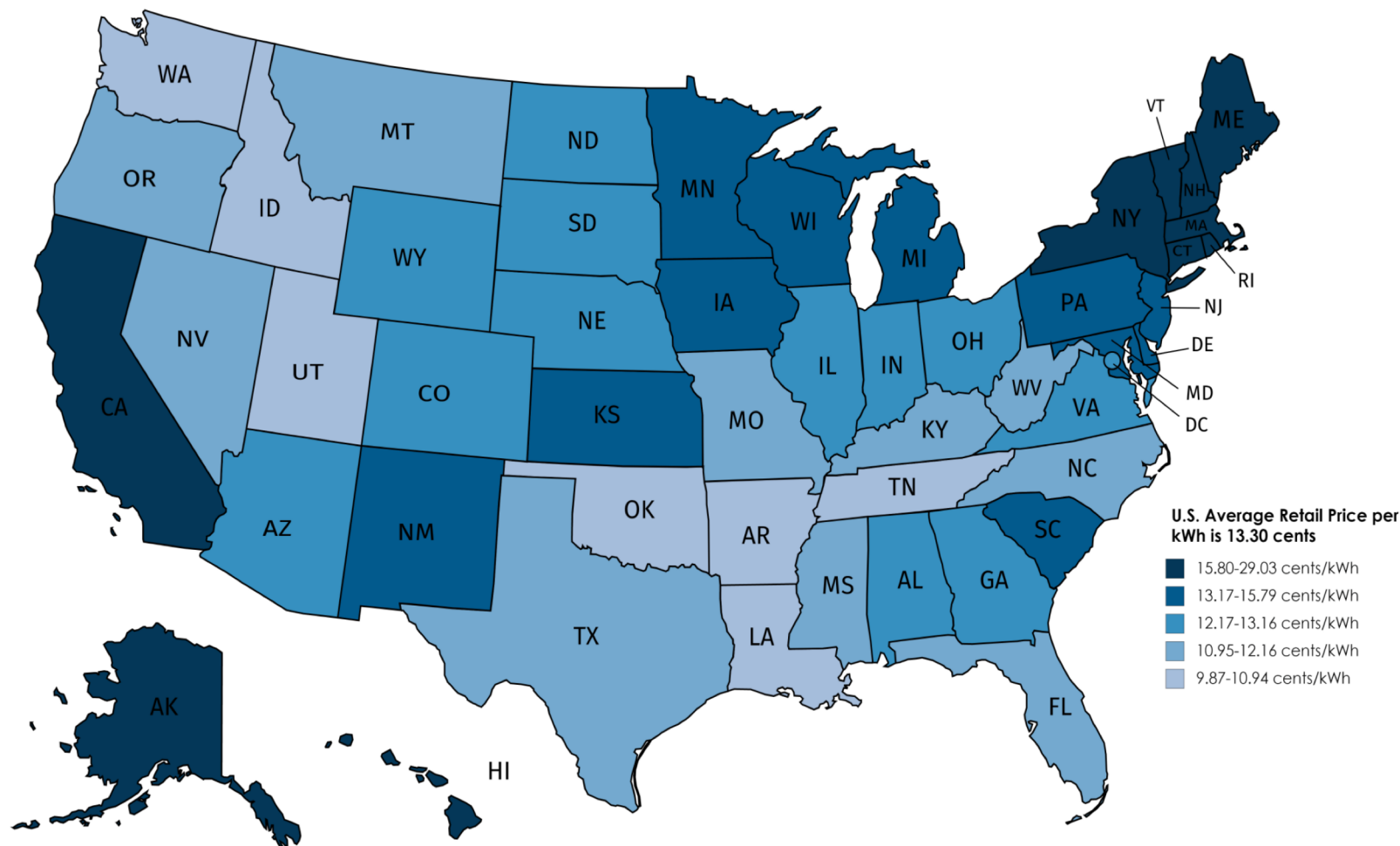
↑ 72%

26%

Sources: Rates (Public Advocates Office analysis of utility data for the residential average rate excluding the California Climate Credit);
Customers Behind on Bills or "Arrearages": (November 2023 utility compliance filings in the CPUC's disconnection proceeding)



Residential Electric Supply Rates in 2018



10 Most Expensive States to Live In

Rank	State	January 2018 Rate (cents per kWh)
1	Hawaii	31.14
2	Rhode Island	22.24
3	Alaska	21.67
4	Massachusetts	20.60
5	Connecticut	20.00
6	New Hampshire	19.23
7	California	18.81
8	New York	17.74
9	Vermont	17.36
10	Maine	16.02

10 Most Cheapest States to Live In

Rank	State	January 2018 Rate (cents per kWh)
1	Louisiana	8.72
2	Oklahoma	8.79
3	North Dakota	9.00
4	Missouri	9.19
5	Nebraska	9.29
6	Arkansas	9.36
7	Washington	9.51
8	Kentucky	9.78
9	Tennessee	10.02
10	North Carolina	10.27